

Task : Linux Permissions

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Course : CyberSecurity

Date: 22-07-26

1. Objective -

The objective of this task was to understand how Linux file and directory permissions work, and to apply that understanding to successfully read the contents of a restricted file `/home/student/practice/restricted_file.txt` on the target machine.

2. Prerequisites – Linux Permissions Model

Every file and directory on a Linux system has an owner, a group, and a set of permissions that control who can **read (r)**, **write (w)**, or **execute (x)** it.

Permissions are split into three categories:

- **User (u)** — the owner of the file
- **Group (g)** — users belonging to the file's group
- **Others (o)** — everyone else on the system

When listed with `ls -la`, permissions appear as `-rw-r--r--`. The first character indicates file type (`-` for a regular file, `d` for a directory) and the remaining characters are three sets of `rwX` for user, group, and others respectively.

3. Step-by-step process

Step 1: Connecting and Confirming Identity

After using SSH to connect to the target machine, I confirmed my session context:

```
student@hacking:~$ pwd
/home/student
student@hacking:~$ whoami
student
```

This confirmed I was logged in as the student user.

Step 2: Checking the home directory

```
student@hacking:~$ ls
loot  practice  read_secret.sh  run_secret
secret.txt
```

This showed two directories (`loot`, `practice`) and three files, including a script (`read_secret.sh`), a binary (`run_secret`), and a plain file (`secret.txt`) which were investigated in the further questions

Step 3: Exploring the `loot` Directory

Basic navigation (`cd`) and file reading (`cat`) confirmed access to this directory's contents

```
student@hacking:~$ cd loot
student@hacking:~/loot$ ls
readme.txt
student@hacking:~/loot$ cat readme.txt
flag{you_know_cd_ls_cat_now}
```

Digging deeper with `ls -la` showed the hidden files, denoted by a leading dot, along with full permission details :

```
student@hacking:~/loot$ ls -la
total 16
drwxrwxr-x 2 student student 4096 Jan 28 06:05 .
drwxr-x--- 6 student perm   4096 Apr 13 11:10 ..
-rw-rw-r-- 1 student student   32 Jan 28
06:01 .hidden_file.txt
-rw-r--r-- 1 student student   29 Jan 28 06:00 readme.txt
```

This revealed a hidden file, `.hidden_file.txt`, which does not appear under a plain `ls`. Since it was owned by `student` with read permission granted to the owner (`rw-`), I was able to read it directly:

```
student@hacking:~/loot$ cat .hidden_file.txt
flag{you_found_the_hidden_file}
```

Step 4: Investigating `restricted_file.txt`

```
student@hacking:~/loot$ cd ..
student@hacking:~$ cd practice
student@hacking:~/practice$ ls
restricted_file.txt
student@hacking:~/practice$ ls -la
total 12
drwxrwxr-x 2 student student 4096 Jan 28 06:05 .
drwxr-x--- 6 student perm    4096 Apr 13 11:10 ..
-r--r--r-- 1 student student   33 Jan 28 06:05
restricted_file.txt
```

The restricted file grants read access to everyone- owner, group, and others alike. It's "restricted" only in the sense that no one, including the owner, has write or execute permission on it.

Since I (logged in as `student`) was the file's owner and owners have read permission here, I was able to read it directly :

```
student@hacking:~/practice$ cat restricted_file.txt
flag{you_understood_permissions}
```

Step 5: Screenshot Evidence

The screenshot shows a Kali Linux terminal window on the left and a quiz interface on the right. The terminal window displays the following commands and output:

```
student@hacking:~$ pwd
/home/student
student@hacking:~$ whoami
student
student@hacking:~$ ls
loot practice read_secret.sh run_secret secret.txt
student@hacking:~$ cd loot
student@hacking:~/loot$ ls
readme.txt
student@hacking:~/loot$ cat readme.txt
flag{you_know_cd_ls_cat_now}
student@hacking:~/loot$ ls -la
total 16
drwxrwxr-x 2 student student 4096 Jan 28 06:05 .
drwxr-x--- 6 student perm    4096 Apr 13 11:10 ..
-rw-r--r-- 1 student student   33 Jan 28 06:01 .hidden_file.txt
-rw-r--r-- 1 student student   33 Jan 28 06:00 readme.txt
student@hacking:~/loot$ cat .hidden_file.txt
flag{you_found_the_hidden_file}
student@hacking:~/loot$ pwd
/home/student/loot
student@hacking:~/loot$ cd ..
student@hacking:~$ cd practice
student@hacking:~/practice$ ls
restricted_file.txt
student@hacking:~/practice$ ls -la
total 12
drwxrwxr-x 2 student student 4096 Jan 28 06:05 .
drwxr-x--- 6 student perm    4096 Apr 13 11:10 ..
-r--r--r-- 1 student student   33 Jan 28 06:05 restricted_file.txt
student@hacking:~/practice$ cat restricted_file.txt
flag{you_understood_permissions}
student@hacking:~/practice$ ls
restricted_file.txt
student@hacking:~/practice$ cd ..
student@hacking:~$ ls
loot practice read_secret.sh run_secret secret.txt
student@hacking:~$ ./read_secret.sh
student@hacking:~$ cat secret.txt
```

The quiz interface on the right contains the following questions and answers:

- What command shows the current directory?
Answer: `pwd` ✓ Correct Answer
- What is the username you are logged in as?
Answer: `student` ✓ Correct Answer
- What is the name of the file located inside the "loot" folder?
Answer: `readme.txt` ✓ Correct Answer
- What are the contents of the file which was in the "loot" folder?
Answer: `flag{you_know_cd_ls_cat_now}` ✓ Correct Answer
- What are the contents of the hidden file in the "loot" folder?
Answer: `flag{you_found_the_hidden_file}` ✓ Correct Answer
- Understand how permissions work in linux and and read the content of `/home/student/practice/restricted_file.txt`
Answer: `flag{you_understood_permissions}` ✓ Correct Answer

The quiz interface also shows a score of 9 out of 9.

4. Conclusion

This task demonstrated core Linux permission concepts in practice: reading permission strings, understanding the owner/group/other model, recognizing hidden files. The objective of reading the `restricted_file.txt` was achieved by correctly interpreting its permission bits (`-r--r--r--`), which confirmed read access was open to all users including the owner.